(a)

Numbering model
Faster R-CNN, 32 FDI classes; best box per code

Tooth detector
one class ‘tooth’; score ≥ 0.3

Segmentation model
Mask R-CNN instance masks

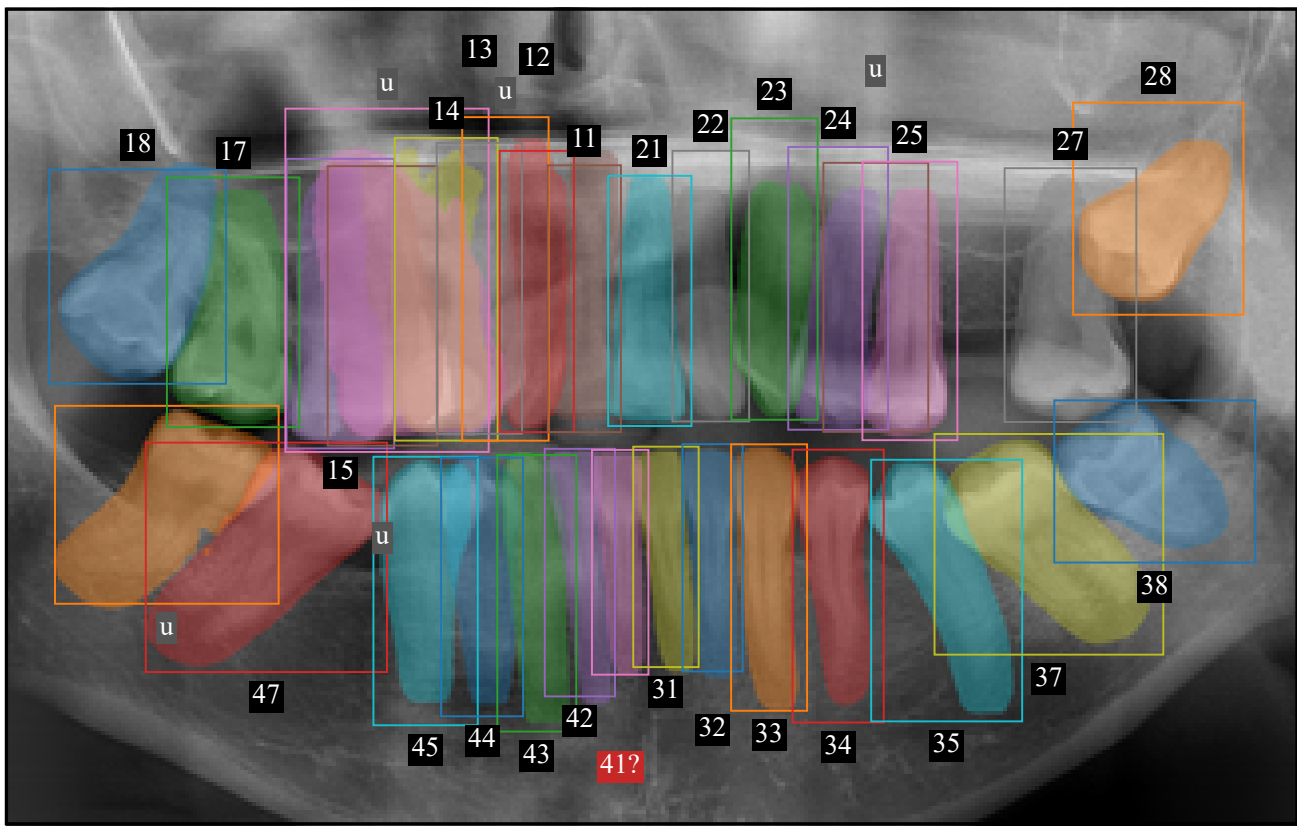
Number assignment
in score order, each box takes the unused numbering box of highest IoU ≥ 0.5; otherwise ‘unassigned’

Mask matching
one-to-one with detector boxes, IoU ≥ 0.5

Consistency check
flag if a code is duplicated, confidence < 0.5, or side / arch disagrees with the quadrant; no number is forced

Region record
tooth box, FDI code or ‘unassigned’, mask, flags

(b)



14 FDI number from the numbering model

41? number flagged for review u no number assigned